IFAS: Intelligent Footfall Analysis System for Retail Environments

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***Abstract*—Most retail stores record many hours of CCTV footage every day, yet very little of it is ever analysed. We present IFAS, a system that processes this footage and reports how many people visited, their gender and approximate age group, how long they stayed and where in the frame they spent their time. Instead of one large model for everything, IFAS uses a separate model for each task. People are detected with YOLOv8 and followed across frames by a simple IoU-based tracker. Gender is predicted by a ViT-B/16 network fine-tuned on the PETA dataset, while age is predicted by four support vector machines trained on colour and texture histograms, in the manner of the original PETA baseline. Because each person appears in many frames, their predictions are pooled into a single label. On PETA the gender model reached up to 97.8% validation accuracy in short training runs, and the age model obtained a mean accuracy of 0.835 over four age groups on a held-out test split (76.7% four-way accuracy), with about four in five of its errors falling in an adjacent age group. Both splits are made at the image level, so the same person can appear on both sides, and these figures are likely to be optimistic. We also study how sensitive each model is to its main settings, measure the runtime of every stage on a laptop CPU, and run the full pipeline on a short shop recording to see where detection and tracking break down and how much two additions, rejoining lost tracks and a second detector for weak detections, are able to repair.**

***Index Terms*—footfall analysis, pedestrian attribute recognition, vision transformer, transfer learning, support vector machine, local binary patterns, retail analytics**

I. Introduction

A store that wants to know how many customers came in on a given day usually has two options: someone at the door with a hand counter, or an overhead sensor at the entrance. Neither tells the store who the visitors were, how long they stayed or which part of the floor they spent time in. The security cameras that are already installed capture all of this, but in practice the recordings are rarely looked at unless something goes wrong.

Going through the footage by hand is not realistic. A person can only watch one screen at a time, and manual counts start to drift after a few minutes. A modest store with eight cameras open for twelve hours generates close to a hundred hours of video every day, so the analysis has to be automated before the footage is of any real use.

In this paper we describe IFAS, the Intelligent Footfall Analysis System. IFAS takes a recorded video as input and produces a record for every person who appears in it, containing their gender, their age group, the times at which they entered and left the frame, and the path they followed. A web dashboard reads these records and presents them as counts, charts and a short written summary.

Our main design decision was to pick, for each sub-task, the simplest model that does the job well. Gender is a binary label with plenty of training data available, which makes a pretrained vision transformer [1] a natural choice. We fine-tune ViT-B/16 in two stages, first on the public PETA dataset [2] and then on footage from the target store. Age is much harder to read from a low-resolution pedestrian crop, and PETA has comparatively few examples of older people. For age we therefore follow the recipe of the original PETA baseline [2] and train one support vector machine per age group on colour and texture histograms. For detection, speed matters more than capacity, so a small YOLOv8 model [3] is used.

The main contributions of this work are:

- an end-to-end pipeline that combines detection, tracking and two attribute classifiers, and pools several predictions per person into one label;
- a sensitivity study of four training settings of the gender model and five feature settings of the age model;
- an evaluation of the age model on a held-out PETA test split, including a look at the kinds of errors it makes;
- stage-by-stage timings on a laptop CPU and the per-person cost that follows from them; and
- tracker extensions that rejoin people the detector has lost and consult a second detector only about weak detections, along with a field check on shop footage showing what each change fixes.

The rest of the paper is organised as follows. Section II reviews related work and Section III describes the system. Section IV presents our experiments, Section V discusses the results and their limitations, and Section VI concludes the paper.

II. Related Work

A. Pedestrian Attribute Recognition

The PEdesTrian Attribute (PETA) dataset was introduced by Deng et al. [2]. It brings together ten smaller surveillance datasets into 19,000 images of 8,705 people, each annotated with 61 binary and four multi-class attributes, gender and age

group among them. Their baseline describes each image with colour and texture features and trains a separate SVM for every attribute. Results are reported as mean accuracy (mA), the average of the accuracy on positive and on negative examples, so a classifier that always predicts the majority class gets no credit for it. Most later work on pedestrian attributes uses the same measure, and so do we. Jia et al. [4] later showed that the usual PETA partition places images of the same person in both the training and the test set, which inflates reported results, and proposed identity-disjoint splits to address this.

B. Vision Transformers

The transformer [5] was originally designed for language. Dosovitskiy et al. [1] applied it to images by cutting an image into 16×16 patches and treating them as a sequence of tokens. When pretrained on a large dataset such as ImageNet-21k, their Vision Transformer (ViT) matches or outperforms convolutional networks, and ViT-B/16 has since become a common starting point for transfer learning. CLIP [6] learns an image encoder jointly with a text encoder. A gender classifier has no use for the text side, so we chose a vision-only backbone.

C. Colour and Texture Descriptors

Local binary patterns (LBP) [7] describe texture by comparing each pixel with a ring of neighbouring pixels. The rotation-invariant uniform variant is unaffected by rotation and by monotonic changes in brightness, and computing it at several radii captures texture at more than one scale. Colour histograms complement this by describing how colours are distributed within a region [8]. Support vector machines [9] work well on fixed-length descriptors of this kind. For histograms the intersection kernel is a popular choice, and it is the kernel used in the PETA baseline [2].

D. Detection, Tracking and Counting

YOLO [10] predicts all the boxes in an image in a single pass, which made real-time detection practical. YOLOv8 [3] is a recent member of the family and includes small models that run on modest hardware. Counting people by detecting them has the advantage of being easy to audit, since every count is backed by a box. Density-map methods estimate crowd size without locating individuals and are better suited to dense crowds, but a store entrance seldom holds more than a few dozen people at once, so detection is enough here. Tracking by detection links boxes across frames, and SORT [11] showed that even a simple association step works well when the detector is reliable.

E. Transfer Learning

A pretrained network is usually adapted to a new task either by linear probing, where only a new output layer is trained on frozen features, or by full fine-tuning, where every weight is updated. Kumar et al. [12] showed that fine-tuning with a randomly initialised head can distort the pretrained features, and that probing first avoids much of this damage. We use this two-phase schedule together with AdamW [13] and a cosine learning rate schedule with warmup [14].

Algorithm 1 Processing one video

Input: frames $f_0, f_1, \ldots$ at r frames per second; stride s; IoU threshold τ; vote gap g; vote cap m
Output: one record per person

$A \leftarrow \emptyset$ ▷ open tracks
for every s-th frame f_k **do**
  $B \leftarrow$ person boxes in f_k with confidence ≥ 0.10
  $P \leftarrow$ pairs of a track in A and a box in B with IoU $\geq \tau$
  for each pair (t, b) in P, highest IoU first **do**
    if t and b are both unmatched **then**
      extend t with b
    end if
  end for
  drop unmatched boxes that lie mostly on a matched person
  rejoin an unmatched box to a recently lost track near it
  start a track for each remaining box, tentative if below 0.4
  ask the second detector about tentative tracks that persist
  close each track unmatched for more than r sampled frames
  for each track t matched in f_k **do**
    if t is confirmed, has fewer than m votes and its last vote is at least g old **then**
      crop t from f_k and classify gender and age
      add both probability vectors to the votes of t
    end if
  end for
end for
discard unconfirmed tracks and tracks seen in fewer than three frames
return label, confidence, dwell time and path of each track

III. System Design

A. Overview

The overall pipeline is shown in Fig. 1, and Algorithm 1 lists the steps in more detail. The video is read frame by frame, a detector finds the people in each frame, and a tracker links the detections that belong to the same person. Crops of every tracked person are passed to the gender and age classifiers, and their outputs are pooled per person. The result is saved as one JSON record per video, which is what the dashboard reads.

B. Detection and Tracking

Detection uses YOLOv8n restricted to the person class. A box needs a confidence of at least 0.4 to start a new track, but weaker boxes down to 0.10 are also kept and are handled as described in Section III-C. Boxes smaller than 8×16 pixels

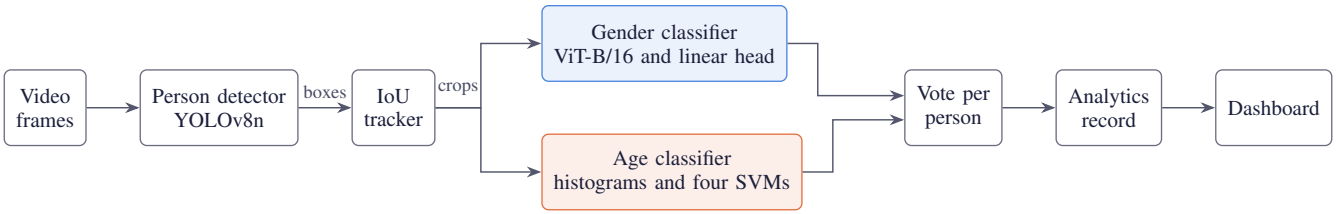


Fig. 1. The IFAS pipeline. People are detected and tracked, each tracked person is classified on several frames by both models, and the pooled labels are written to one record per video.

are thrown away because they hold too little detail to classify, and detection is run on every second frame, which halves its cost.

Tracking is done by greedy matching on intersection over union (IoU), that is, the area two boxes share divided by the total area they cover. Every pair of an open track and a new box with an IoU of 0.3 or more is a candidate. Candidates are taken from the highest IoU downwards, and a pair is accepted only if neither the track nor the box has been matched already. A track is closed once it has gone unmatched for as many sampled frames as the video has frames per second, which works out to two seconds at the default stride. Tracks seen in fewer than three frames are assumed to be detector noise and are dropped.

Two further rules stop one person from being counted several times. The first discards a box when half or more of it lies on a person who has already been matched in that frame. This removes double detections, for example a person and their reflection in a glass panel. The second rule rejoins people who have been lost. People rarely leave the scene from the middle of the frame, so a track that ends there has most likely been dropped by the detector. If a box with a confidence of at least 0.25 then appears within one body height of that spot within the next 8 s, it continues the old track. Near the edges of the frame, where people really do leave and arrive, a box rejoins a track only if the earlier person was standing still and the new box is in the same place.

The tracker has no appearance model. This keeps it cheap, but it also has clear weaknesses, which are listed in Section V-E.

C. Weak Detections and the Second Detector

As mentioned above, the detector runs with a confidence floor of 0.10, well below the 0.4 needed to start a track. Boxes that fall in between are used in two ways. A weak box that overlaps someone who is already being tracked simply extends that track, since a weak detection on a known person is usually genuine. A weak box anywhere else opens a tentative track, which is followed like any other track but is not counted.

A tentative track can become a person in two ways. If a later box on the same track reaches 0.4, the track is confirmed. Otherwise, once the track has collected eight detections, a second and larger detector (YOLOv8m) is run on the same frame at input sizes of 640 and 960 pixels, and the track is confirmed if it finds a person on that spot with a confidence of 0.25 or more. This check is made at most four times per track and no more than once a second. A track that is never confirmed is discarded, and its location is remembered so that the same object is not checked again.

Once confirmed, a track keeps its full history, so the dwell time of that person is counted from the first weak sighting. If the track started where a known person was recently lost, the two tracks are merged.

The main reason for this arrangement is cost. As Section IV-G shows, the larger model finds people that the small one misses but is roughly six to eleven times slower. Running it only on tentative tracks gives most of its benefit for a small fraction of the cost.

D. Gender Classifier

Each crop is resized to 224×224 pixels and normalised with the ImageNet mean and standard deviation. The backbone is the public ViT-B/16 checkpoint `google/vit-base-patch16-224`, which has 86 million parameters and was pretrained on ImageNet-21k and then fine-tuned on ImageNet-1k. It splits the crop into 196 patches, adds a class token and passes the sequence through 12 encoder layers. A pooling layer maps the class token to a 768-dimensional vector, and a single linear layer with 1,538 parameters maps that vector to two logits, which a softmax turns into probabilities (Fig. 3a). The loss is cross-entropy.

One detail of this setup deserves mention. The public checkpoint does not include weights for the pooling layer, so that layer starts from random values. In our code it belongs to the backbone and so stays frozen during the linear probe described below, which means the probe is trained on a random projection of the class token. This may partly explain why the probe on its own stops at about 0.79 accuracy (Section IV-B).

Training runs in two stages, each with two phases (Table I). In Phase A, the linear probe, the backbone is frozen and only the new head is trained, with a learning rate of 10^{-3}. In Phase B the backbone is unfrozen and the whole network is trained with a learning rate of 10^{-5}. Stage 1 runs both phases on PETA. Stage 2 repeats them on labelled crops from the target store, with both learning rates divided by ten so that the model can adapt to the store without forgetting what it learned from PETA. Stage 2 is part of the training code, but we do not evaluate it in this paper.

Within each phase the learning rate rises linearly from zero over the first part of the phase, a fraction set by the warmup

TABLE I
TRAINING SCHEDULE OF THE GENDER MODEL, WITH THE SETTINGS CHOSEN IN SECTION IV-B

Stage	Phase	Trained	Epochs	Peak rate
1, PETA	A, probe	head	4	10^{-3}
	B, fine-tune	whole network	10	10^{-5}
2, store	A, probe	head	2	10^{-4}
	B, fine-tune	whole network	5	10^{-6}

AdamW, batch size 64, weight decay 0.3, warmup ratio 0.2, gradient clipping at 1.0, mixed precision

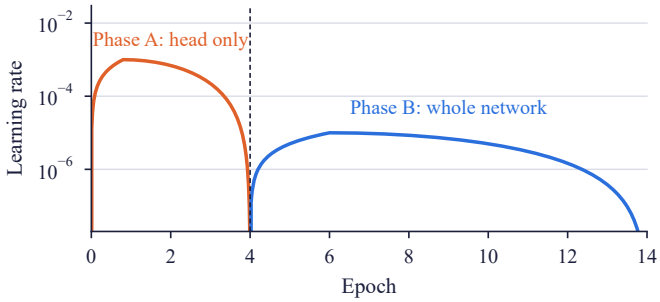


Fig. 2. Learning rate over Stage 1 with the chosen settings. Each phase warms up over the first fifth of its steps and then follows a cosine decay. Note that the vertical axis is logarithmic.

ratio, and then decays to zero along a cosine curve (Fig. 2). All runs use AdamW [13] with a batch size of 64, mixed precision and gradient clipping at 1.0. Training images are flipped horizontally with probability 0.5 and their brightness, contrast and saturation are jittered slightly. One tenth of the data is held out for validation, using a split stratified by gender.

E. Age Classifier

The age model does not use a neural network at all. Each crop is cut into R horizontal strips of equal height; with four strips these correspond roughly to the head, the torso, the upper legs and the lower legs. Two kinds of histogram are computed for every strip. For colour there is one histogram of B bins for each of the red, green, blue, hue and saturation channels. For texture there is one histogram of rotation-invariant uniform LBP codes for each radius in a set $\mathcal{S}$, computed with Q sampling points, which gives $Q + 2$ bins per radius. All histograms are L1-normalised and the histograms of all strips are concatenated into one feature vector (Fig. 3b) of length

$$d = R\left(5B + |\mathcal{S}|\,(Q+2)\right). \tag{1}$$

We train one SVM for each of the four age groups, 16 to 30, 31 to 45, 46 to 60 and above 60. Each one is a binary classifier that separates its own group from the other three. For every group we try two kernels: the histogram intersection kernel

$$K(\mathbf{x},\mathbf{y}) = \sum_i \min(x_i, y_i), \tag{2}$$

and the standard radial basis function (RBF) kernel with width parameter γ. The penalty C is searched over $\{0.01, 0.1, 1, 10, 100\}$ for both kernels and γ over $\{0.001, 0.01, 0.1, 1\}$ for the RBF kernel, which gives 25 settings per group, and the one with the highest validation mean accuracy is kept. For a single attribute, mean accuracy is

$$\mathrm{mA} = \frac{1}{2}\left(\frac{TP}{P} + \frac{TN}{N}\right), \tag{3}$$

where P and N are the numbers of positive and negative examples and TP and TN are how many of each are classified correctly. At test time a crop is given the age group whose SVM returns the largest decision value.

PETA also has a label for people under 15, but there are too few such images to train a stable classifier, and our early models underfitted this group every time. We therefore left it out.

F. Pooling Predictions per Person

Any single crop may be blurred or partly hidden, but a person is usually visible for many frames, so IFAS classifies each track several times and combines the answers. A track is classified at most once every 0.4 s and at most 12 times in total. Spacing the samples out avoids classifying near-identical crops, and the cap limits the cost of someone who stays in view for several minutes.

Each classified crop votes for the label it finds most probable, and the track receives the label with the most votes, with ties going to the label with the higher total probability. The confidence we report for a person is simply the average probability that the crops gave to the winning label. For the age model, the SVM decision values are first turned into a distribution with a softmax. This puts age confidence in the same 0 to 1 range as gender confidence, but the two are not calibrated against each other and should not be compared directly.

G. Analytics and Dashboard

For every track IFAS records when the person was first and last seen and the dwell time in between, and sorts the dwell time into one of the five tiers in Table II. The video is also divided into 5 s bins, and for each bin the record stores the number of arrivals and exits along with the mean and peak number of people in view. A grid of 16×9 cells laid over the frame accumulates where people stood, and the frame is further divided into three vertical zones.

Since recordings vary in length, IFAS compares them by rate rather than by raw count. Each recording is split by the clock hours it covers, and for every hour we store the number

TABLE II
DWELL TIERS

Tier	Time in frame	Reading
Passing	under 3 s	walked through
Glancing	3 to 10 s	slowed or looked
Browsing	10 to 30 s	stopped
Engaged	30 to 60 s	stayed
Dwelling	over 60 s	waited or queued

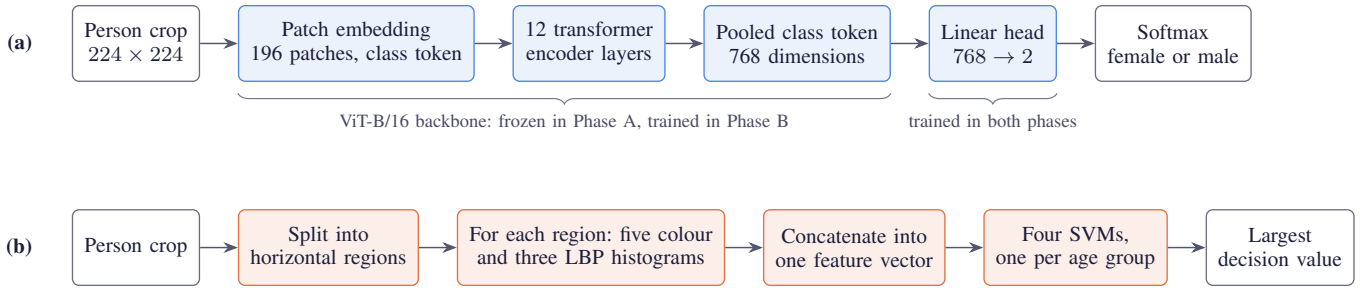


Fig. 3. The two attribute classifiers. (a) The gender network, a pretrained ViT-B/16 with a new linear head. (b) The age model, in which region histograms are scored by one SVM for each age group.

of arrivals and the number of minutes that were actually watched. The arrival rate for any group of hours, such as all Saturdays, is the total number of arrivals in those hours divided by the total minutes watched, expressed per hour. This way a three-minute clip cannot outweigh a three-hour export. A day is only included if it has at least five minutes of footage, and a trend is drawn only once three such days are available.

The dashboard shows the results at two levels of detail. The simple view, meant for the store manager, summarises the findings in plain sentences alongside a few headline charts. The detailed view adds filters by gender, age group, dwell tier and confidence, a table of every tracked person and the confidence distributions of both models.

IV. Experiments

A. Data

All experiments use PETA [2]. For the gender model we use every image that has a gender label, about 19,000 in all, and hold out one tenth for validation.

The age model uses the 14,241 images labelled with one of the four age groups we keep. Fig. 4 shows how they are distributed: the two youngest groups hold 79% of the images and the oldest only 8%. We split them into training, validation and test sets in the same 9,500 : 1,900 : 7,600 proportions as the PETA protocol, giving 7,120 training, 1,424 validation and 5,697 test images, stratified by age group.

Both splits are made at the image level. PETA contains several images of many of its 8,705 people, so the same person can end up in the training data and in the validation or test data. This is the same problem Jia et al. [4] found in the standard PETA partition, and it means the accuracies reported in this section are probably higher than what the models would achieve on people they have never seen, such as new customers in a store.

B. Gender Model

We varied four Stage 1 settings one at a time: the backbone learning rate, the number of linear probe epochs, the weight decay and the warmup ratio. Table III lists the values tried and Fig. 5 shows the results. To keep the study affordable the runs were short, between three and seven epochs in total, and each setting was trained only once. With about 1,900 validation images and accuracy near 97.5%, the standard error

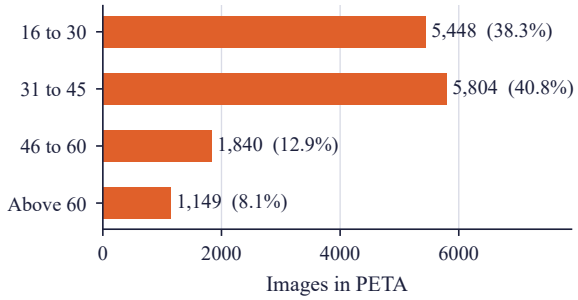


Fig. 4. Images per age group among the 14,241 PETA images used for the age model.

TABLE III
Stage 1 settings tested for the gender model

Setting	Values tested	Selected
Backbone learning rate	10^{-4}, 10^{-5}, 10^{-7}	10^{-5}
Linear probe epochs	0, 1, 2, 3, 4	4
Weight decay	0, 0.001, 0.01, 0.05, 0.1, 0.3, 0.5	0.3
Warmup ratio	0, 0.05, 0.1, 0.2, 0.3, 0.4	0.2

of a single validation accuracy is roughly 0.36 points, and differences smaller than that should be treated as noise.

Learning rate. With a backbone learning rate of 10^{-7} the model hardly changes, and validation accuracy stays around 0.80, roughly what the linear probe reaches on its own. Rates of 10^{-4} and 10^{-5} both converge, reaching 0.992 and 0.983 after four epochs of fine-tuning (Fig. 5a). The higher rate did a little better in this short run, but we chose 10^{-5} because a smaller step is less likely to damage the pretrained features over the longer schedule of the final model. This is a judgement on our part rather than something the experiment shows. These runs also fine-tuned for one epoch longer than the probing and weight decay runs, so their numbers cannot be compared directly with the other panels.

Linear probe. Skipping the probe gives the lowest accuracy, 96.8%. Any amount of probing helps, and four epochs gave the best result at 97.8% (Fig. 5b). The trend is not monotonic, since two epochs scored lower than one, which we put down to run-to-run noise. Interestingly, the training loss tells the opposite story: over the first 200 steps of fine-tuning, the loss

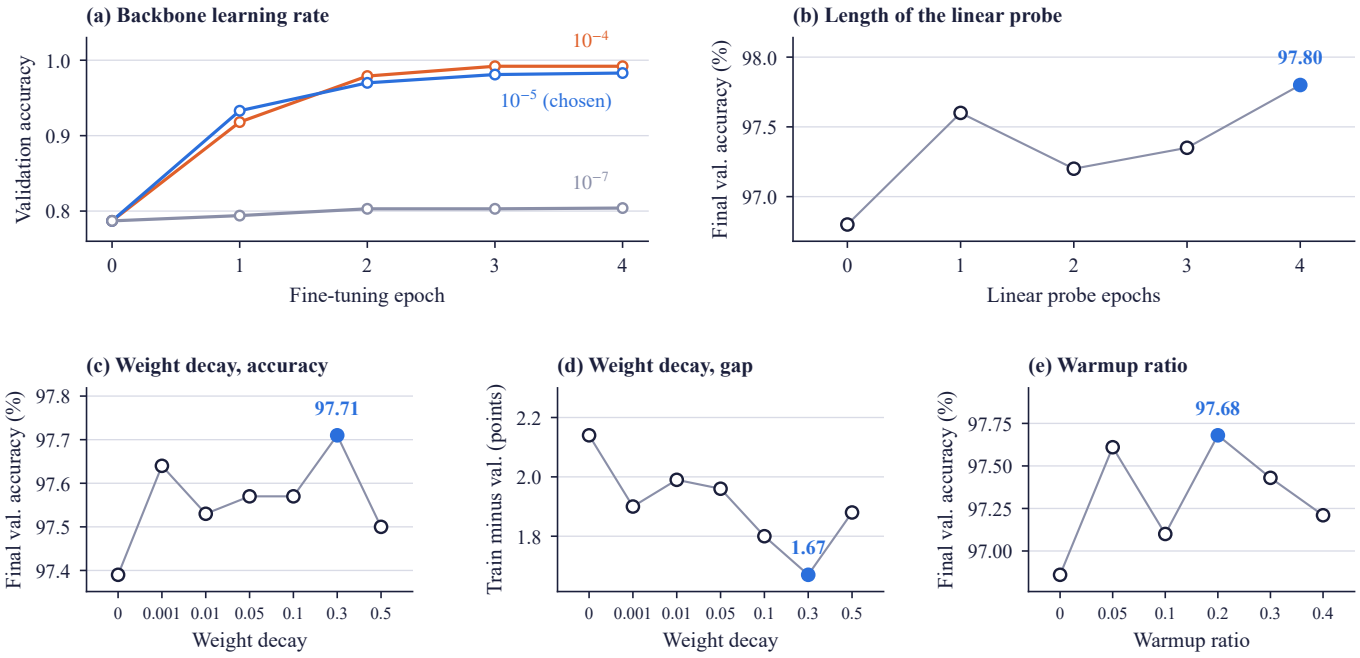


Fig. 5. Sensitivity of the gender model to four training settings on PETA. The selected value is filled in blue. (a) Validation accuracy during fine-tuning for three backbone learning rates; epoch 0 is the end of the linear probe. (b) to (e) Final validation accuracy, or the gap between training and validation accuracy, with one setting varied and the others fixed. Each point is a single run.

falls further without a probe than with one. Lower training loss together with lower validation accuracy is what we would expect if the network were fitting the training set at the cost of its pretrained features [12].

Weight decay. Accuracy was lowest without weight decay (97.39%) and highest at a decay of 0.3 (97.71%), and the gap between training and validation accuracy was also smallest at 0.3, at 1.67 points compared with 2.14 points with no decay (Fig. 5c and d). However, the whole range across the seven values is only about a third of a point, which is within the noise of a single run. The study does not really separate these values, and we kept 0.3 mainly because it came out on top in both measures.

Warmup. Training without warmup gave the lowest accuracy, 96.86%. A ratio of 0.2 gave the highest at 97.68%, with 0.05 close behind at 97.61% (Fig. 5e). The non-zero ratios follow no clear order.

Overall, the probe changes accuracy by 1.0 point across its range, warmup by 0.8 and weight decay by 0.3. Only the first two are clearly larger than the noise level. We should also be clear about where the 97.8% figure in the abstract comes from. It is the best run of the probe study (four probe epochs followed by three epochs of fine-tuning), not a run of the full schedule in Table I. It was also picked on the same validation set it is reported on, and it is affected by the identity overlap described in Section IV-A, so it is best read as an upper estimate.

TABLE IV
FEATURE SETTINGS TESTED FOR THE AGE MODEL

Setting	Values tested	Selected
Body regions R	2, 3, 4, 5, 6, 7	4
Colour bins B	8, 16, 24, 32	16
LBP points Q	8, 16, 24	16
LBP radii $\mathcal{S}$	{1}, {1,2}, {1,2,3}, {1,2,3,4}, {2,3}	{1,2,3}
Class weight	none, balanced	balanced

C. Age Model: Feature Settings

To keep the search fast, the feature settings were screened on a 30% subsample of the training and validation data. We tuned them in the order listed in Table IV, fixing each choice before moving on to the next, so interactions between settings were not explored. Fig. 6 shows the average mA over the four age groups.

The number of body regions has the largest effect. Average mA rises from 0.7155 with two regions to 0.7457 with four and then drops off slightly (Fig. 6a). Our explanation is that strips thinner than a quarter of the body contain too few pixels for a stable histogram. Colour bins behave in a similar way: 16 bins score 0.7457, while 32 bins fall to 0.7220 as the histograms become sparse (Fig. 6b). The number of LBP sampling points matters less, with 16 points doing best at 0.7493 (Fig. 6c). The five combinations of LBP radii we tried all scored within a narrow band, with radii of 1, 2 and 3 marginally ahead. Finally, balanced class weights raised average mA from 0.7432 to 0.7493 (Fig. 6d), which

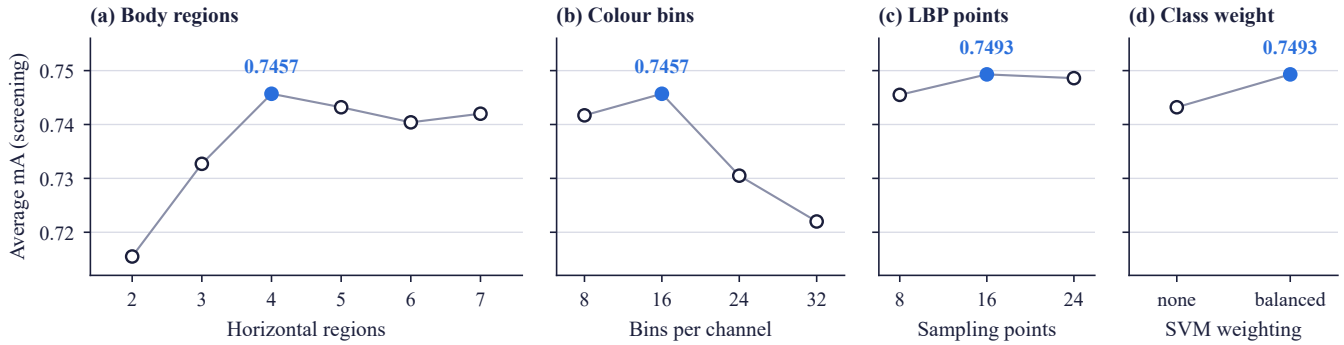


Fig. 6. Sensitivity of the age model to its feature settings, measured as average mA on a 30% subsample of PETA. All four panels share one vertical scale. The selected value is filled in blue.

TABLE V
FINAL AGE CLASSIFIERS: KERNEL SETTINGS AND VALIDATION MA

	Released, $d = 550$			Selected, $d = 536$		
Age group	C	γ	mA	C	γ	mA
16 to 30	1	1	0.7996	1	1	0.8005
31 to 45	10	1	0.7823	1	1	0.7829
46 to 60	10	0.1	0.7897	10	0.1	0.7866
Above 60	10	0.1	0.9108	10	0.1	0.9204
Mean			0.8206			0.8226

TABLE VI
RELEASED AGE MODEL ON THE PETA TEST SPLIT. PRECISION, RECALL AND F1 TREAT THE FOUR SVMS AS ONE FOUR-WAY CLASSIFIER; MA USES EACH SVM'S OWN ONE-VS-REST DECISIONS.

Age group	Images	Precision	Recall	F1	mA
16 to 30	2,179	0.730	0.812	0.769	0.8107
31 to 45	2,322	0.794	0.720	0.755	0.7888
46 to 60	736	0.730	0.728	0.729	0.8312
Above 60	460	0.897	0.848	0.872	0.9102
All	5,697	accuracy 0.767, macro F1 0.781			0.8352

is expected given the imbalance in Fig. 4.

D. Age Model: Final Classifiers

We trained the final SVMs on the full training set in two configurations (Table V). The first is the configuration released with the pipeline. It uses five regions and eight LBP points, which by (1) gives 550 features per crop. The second uses the settings selected above, four regions and 16 points, for 536 features. In both cases the grid search picked the RBF kernel for every age group, so neither configuration ends up using the intersection kernel of the PETA baseline. On the validation split the two are very close, with mean mA of 0.821 and 0.823, and the selected configuration does slightly better on the oldest group (0.920 against 0.911).

E. Age Model: Test Results

The released classifiers were evaluated once on the 5,697 held-out test images, and Table VI and Fig. 7 summarise the results. Mean accuracy averaged over the four groups is 0.8352, a little above the validation figure of 0.8206. The biggest change is in the 46 to 60 group, which goes from 0.790 to 0.831. The validation set holds only about 180 images from this group, so we read the rise as sampling variation rather than a real improvement.

When the four SVMs are used together as one four-way classifier, 76.7% of test images are labelled correctly and the macro-averaged F1 score is 0.78. The oldest group is the easiest, with a precision of 0.90 and a recall of 0.85, while the 46 to 60 group is the hardest, with an F1 of 0.73.

The confusion matrix in Fig. 7a shows where the errors go. Of the 1,330 misclassified images, 1,057 (79.5%) are put in a group next to the true one. If such near misses are accepted, 95.2% of test images land in the correct group or a neighbouring one. The most common confusion is between the two youngest groups: 23% of people aged 31 to 45 are labelled 16 to 30, and 14% are confused the other way round.

F. Runtime

We timed each stage on a laptop with a two-core AMD Ryzen 3 3250U processor and no dedicated graphics card. The input was an 848×478 recording, and the classifiers were timed on the person crops the detector found in it. Median times are shown in Fig. 8. Detection takes 157 ms per frame. The age model needs 55 ms per crop, of which 43 ms goes into computing the histograms. The gender network takes 794 ms per crop in batches of 16 and 900 ms for a single crop, and it accounts for most of the total cost, which is why the number of votes is capped. With at most 12 votes, the two classifiers cost under 12 s per person on this machine however long the person stays. Without the cap, a person would be classified up to two and a half times for every second they spend in view, so a customer who stayed for a minute would need about 150 classifications and nearly two and a half minutes of computation. On a processor of this class IFAS is therefore a tool for recorded footage, and a live feed would need faster hardware.

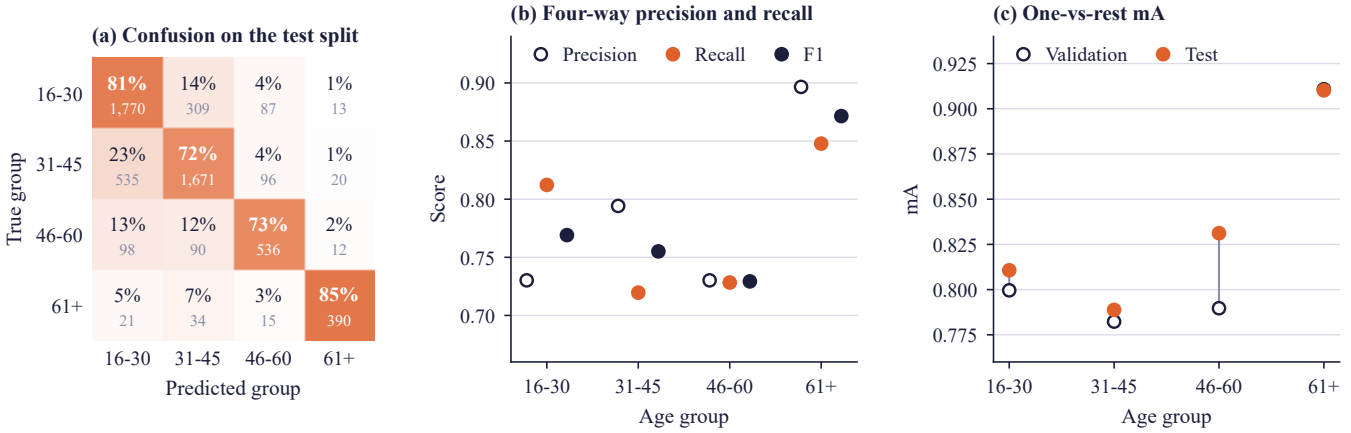


Fig. 7. The released age model on the PETA test split. (a) Confusion matrix. Each row sums to 100%, and the small figure is the number of images. (b) Precision, recall and F1 when the four SVMs act as one four-way classifier. (c) One-vs-rest mean accuracy on the validation and test splits.

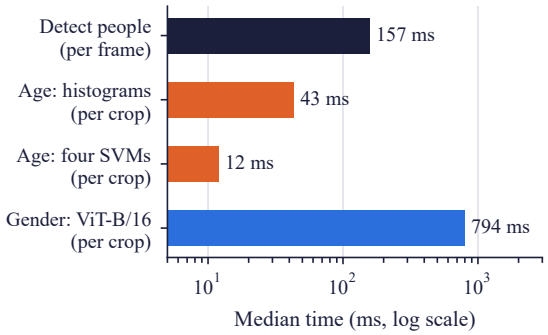


Fig. 8. Median time per stage on a two-core laptop processor. The horizontal axis is logarithmic. Detection is per frame and the classifiers are per person crop.

TABLE VII
SHARE OF 32 FRAMES IN WHICH EACH PERSON IS DETECTED AT A CONFIDENCE OF 0.25 OR MORE

Detector, input size	ms	Seated	Walking	Left desk	Behind glass
YOLOv8n, 640	183	100%	50%	38%	0%
YOLOv8s, 640	433	100%	62%	6%	0%
YOLOv8s, 960	890	100%	53%	3%	12%
YOLOv8m, 640	1,057	100%	72%	66%	0%
YOLOv8m, 960	2,084	100%	75%	25%	44%
YOLOv8n, 640, at 0.05	183	100%	81%	100%	100%

ms is the time per frame on the laptop of Section IV-F. The last row also produced 50 boxes that lie on no person.

G. Field Check on Shop Footage

PETA only measures the classifiers and tells us nothing about detection or tracking. To get some sense of these, we ran the whole pipeline on a recording from outside the dataset: a 15 s clip of a shop interior at 1280×720 pixels, with glass partitions and boxes stacked around the room. We annotated the clip by eye. Five people are in view for the whole clip: a man seated in the open, a second man who walks across the room and then works behind a glass partition, a person sitting at a desk on the left edge of the frame, a person seated behind glass at the back, and a fifth person standing almost fully hidden behind the third.

Detection. Table VII shows how often four of these people are detected in 32 evenly spaced frames. The fifth is never detected as a separate person under any setting, and no setting finds everyone. At its normal threshold the small model never finds the person behind glass. The medium model at 960 pixels finds that person in 44% of frames, but then picks up the person at the left desk in only 25% of them. Dropping the small model's threshold to 0.05 finds all four people, but it also fires on a display stand, and with higher confidence than it gives the person behind glass, so a lower threshold alone cannot tell the two apart. The YOLOv8n time in this table (183 ms) differs from the 157 ms of Section IV-F because it was measured on a different clip in a separate run.

Tracking. Table VIII compares two versions of the pipeline on this clip. The first uses IoU matching alone with a single confidence threshold of 0.4, and the second adds the rules from Sections III-B and III-C. The first version reports five visitors. This happens to be the true number, but the details behind it are all wrong. The walking man is counted three times, because the detector loses him behind the glass, and two people are missed altogether. The mean dwell time comes out at 4.3 s for people who were present for the full 15 s, and three of the five tracks are classed as passing through.

The second version reports four visitors, each of them a different real person. The walking man is followed as a single track, the person at the left desk is tracked for the whole clip, and the person behind glass is confirmed by the second detector. The display stand opens a tentative track twice and is rejected both times. The second detector was consulted on 11 of the 188 sampled frames, and the processing time on the laptop went up from 77 s to 205 s.

TABLE VIII
THE PIPELINE ON THE SHOP CLIP, BEFORE AND AFTER THE TRACKING RULES. EACH CELL IS THE DWELL TIME REPORTED FOR THAT PERSON.

Person (present for 15 s)	IoU matching only	With all rules
Seated in the open	15.0 s	15.0 s
Walking, then behind glass	3.4, 1.5 and 1.0 s	14.7 s
At the left desk	0.6 s	15.0 s
Behind glass at the back	missed	11.6 s
Hidden behind another	missed	missed
Tracks reported	5	4
Distinct people found	3 of 5	4 of 5
Mean dwell time	4.3 s	14.1 s

We stress that this is a single clip, chosen because it is difficult, and that the tracking rules were tuned on this same clip. The comparison shows what kind of error the basic version makes and what kind of rule fixes it, but it is not a measurement of accuracy. The classifiers also did worse here than on PETA. Gender was stable for the three people who are clearly visible, but for the person at the left desk it was close to a coin toss, with a confidence of 0.56. Age confidence stayed below 0.45 for everyone. Part of this is due to the softmax over SVM decision values, which can give a fairly flat distribution even when one SVM is clearly ahead, so these values should not be read as calibrated probabilities.

V. DISCUSSION

A. Staged Transfer Learning

Our probing result is in line with Kumar et al. [12]. A newly added head starts from random weights, so its first gradients are large and carry little useful information, and if the backbone is trainable at that point these gradients flow straight into it. Training the head alone for a few epochs first avoids the problem at little extra cost. In our case the randomly initialised pooling layer described in Section III-D adds a second set of random weights, and moving it into the head, or reading the class token directly, is worth testing.

The weight decay of 0.3 that came out best is higher than the 0.01 to 0.1 usually used with AdamW. With about 19,000 images and 86 million parameters, it is plausible that stronger regularisation helps, but as noted in Section IV-B the measured differences are within single-run noise. Repeating the study with several random seeds is the first thing we would do with more compute.

B. Reading the Age Results

The oldest group is the easiest on every measure. Grey or white hair changes the colour histogram of the top region, and posture or clothing may also contribute, although this is only our interpretation and we did not test which features are actually responsible.

The two youngest groups are confused with each other more than any other pair, which is not very surprising, since a 29-year-old and a 32-year-old fall on opposite sides of a boundary but look much the same at surveillance resolution. The confusion matrix supports the view that most errors are boundary effects: four errors in five are off by a single group, and errors of two or three groups make up only 4.8% of the test set. For a store that wants a rough age profile of its visitors, being one group out matters far less than being three groups out. Mean accuracy does not reward this, whereas a measure that respects the order of the groups would.

Even though the oldest group has only 8% of the data, it reaches a recall of 0.85, which suggests that the balanced class weights are doing their job of pushing each SVM to find its positive examples.

Leaving out the under-15 group points to a wider problem. Surveillance datasets contain few children, partly because few appear in such footage and partly because privacy rules limit how this kind of data can be collected. A store that needs to know how many children visit should not rely on IFAS for it.

Finally, although our age model follows the recipe of the PETA baseline, its numbers cannot be set directly against those of Deng et al. [2]. We use our own image-level split of a subset of PETA, we drop one age group, and the grid search chose the RBF kernel instead of the intersection kernel. A proper comparison would need the original partition and the intersection kernel.

C. Feature Settings and the Released Model

The released model and the selected settings differ in the number of regions and the number of LBP points. On validation data their mean mA differs by only 0.002, which is too small to separate them on a single split. The screening study is still useful, because it shows that the model is not fragile. Across all the settings we tried, average mA stays between 0.7155 and 0.7493, a spread of 3.4 points, and the two choices that hurt most are too few body regions and too many colour bins.

D. Why Two Kinds of Model

A single network predicting both gender and age would be simpler to deploy, but we kept the two separate for two reasons. The age model follows a published baseline, and its features can be inspected directly, since each histogram bin is a colour range in a named part of the body. The gender task, on the other hand, has enough data to train a large network, and the accuracy of the transformer on it is hard to match with hand-crafted features.

The runtime measurements give a third reason. Per crop, the age model costs less than a tenth of the gender network on a CPU, so a store without a graphics card can afford to run age on every crop and spend most of its compute budget on gender.

E. Limitations

Apart from the field check, every number in this paper comes from PETA, and the field check itself is a single 15 s clip annotated by eye. Accuracy in a real store therefore remains unmeasured. Several other limitations follow from the design.

- Our PETA splits are made at the image level, so some people appear on both sides. The reported accuracies are likely to be optimistic for unseen customers, and an identity-disjoint split such as that of Jia et al. [4] should be used in future work.
- The gender results are validation accuracies from short single runs, with the best one chosen on the same validation set. The gender model has no held-out test result yet, and Stage 2 has not been evaluated.
- The tracker does not re-identify people by appearance. Someone who leaves the frame and comes back is counted twice, and two people who cross paths can swap tracks. The rejoining rule is a heuristic and can merge two people who stand in the same spot a few seconds apart.
- A person whom no detector finds is never counted. In the field check, one of the five people, hidden behind another, was missed for the entire clip.
- The tracking rules were tuned on a single recording and still need to be checked on labelled footage from other stores.
- Occupancy is the number of people in view, not the number in the store, and dwell time measures presence in the frame, which is only a proxy for interest.
- The age SVMs were trained at PETA resolution, so small or blurred crops receive low-confidence labels, and age confidence is not calibrated.
- Per-track accuracy has not been measured. Pooling helps most when the errors on different frames are independent, but crops of one person taken a fraction of a second apart share lighting and pose, so their errors are likely to be correlated.
- The runtime figures come from one processor and one recording.

VI. Conclusion

In this paper we presented IFAS, a system that turns store footage into a record of who came in, when they came and how long they stayed. It combines models chosen task by task: a small detector for speed, a fine-tuned transformer for gender and an SVM baseline for age. On PETA the gender model reaches up to 97.8% validation accuracy, and the age model achieves a mean mA of 0.835 on the test split, with 95% of its predictions within one age group of the truth, although both figures come from image-level splits and are probably optimistic. The sensitivity study shows that linear probing before fine-tuning has the largest effect on the gender model, and that the number of body regions matters most for the age model. The timings show that the gender network dominates the cost and that capping the votes per person keeps that cost bounded. On a difficult clip from outside PETA, rejoining lost tracks and checking weak detections with a second detector raised the number of people found from three of five to four of five and brought the reported dwell times in line with the footage, although the rules were tuned on that same clip.

We see three directions for future work. The first is to label footage from real stores and measure per-track accuracy on it, using identity-disjoint splits for any retraining. The second is to add re-identification so that returning visitors are counted only once. The third is to run the pipeline on a live camera feed, which we have not tested yet.

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